SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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Abstract.

Contents

1. Introduction

1.1. Background and motivation.

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$ and let G be the corresponding finite reflection group. Let $k : R \to \mathbb{R}_+$ be G-invariant multiplicity function. Let $R_{\mathrm{Dunkl},j}$ be the j-th Dunkl-Riesz transform corresponding to R and k. Let w be the Dunkl weight corresponding to R and k. Let $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$.*

(i) *If $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, then $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \le c_{p,R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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(ii) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, $p > N$, *then* $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{p,R,k}^{-1}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

(iii) *If* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, *then* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{R,k}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

(iv) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, *then* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \geq c_{R,k}^{-1}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

1.2. **Strategy of the proof.** In Section 3, we introduce our *ad hoc* Sobolev space and demonstrate that its semi-norm is controlled by the usual Sobolev semi-norm. In Section 4, we use Janson-Wolff method to control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ by the *ad hoc* Sobolev semi-norm of f. Thus, we control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ by the usual Sobolev semi-norm (i.e., we prove Theorem 1.1 (i)).

In Section 5, we control the oscillation semi-norm of f (with respect to measure $w(x)dx$ and distance $(x, y) \to |x-y|_\infty$) via $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$. After that, we verify the assumptions in Proposition 1.2 in [6] for measures $w(x)dx$ and dx and, using the latter proposition, conclude that oscillation semi-norm of f with respect to measure $w(x)dx$ and distance $(x, y) \to |x-y|_\infty$ is equivalent to the one with respect to measure dx and distance $(x, y) \to |x-y|_\infty$. The latter is known to be equivalent to the usual Sobolev semi-norm of f.

1.3. **Notation.** Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N, p, and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda\,\ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p, which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ-measurable set E by

$$(f)_E := \fint_E f(x)d\mu(x) := \frac{1}{\mu(E)}\int_E f(x)d\mu(x). \tag{1.1}$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. **Harmonic analysis in the Dunkl setting.** Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle\cdot, \cdot\rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v\rangle v.$$

A finite set $R \subset \mathbb{R}^N \backslash \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $|v| = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v\in R}$ is called the Weyl group (reflection group) of the root system R. Let $k : R \to [0, \infty)$ be a multiplicity function, which is a G-invariant function fixed throughout the paper.

Define the Dunkl measure $w(x)dx$ on $\mathbb{R}^N$ by

$$w(x) = \prod_{v\in R} |\langle x, v\rangle|^{k(v)}.$$

Let $B(x, r) := \{y \in \mathbb{R}^N : \ |x - y| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Furthermore,

$$w(B(x, r)) \sim r^N \prod_{v\in R} (|\langle x, v\rangle| + r)^{k(v)}, \tag{2.1}$$

where the implicit constant only depends on N, R and k. It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$C^{-1}\left(\frac{r_2}{r_1}\right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C\left(\frac{r_2}{r_1}\right)^{N'}, \text{ for } 0 < r_1 < r_2, \tag{2.2}$$

so the numbers N' and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v\in R} \frac{k(v)}{2}\langle v, \xi\rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x\rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ. Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = -\sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}\Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \cdots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$\left\{x \in \mathbb{R}^N : \langle x, v\rangle \neq 0 \ \ for\ all\ v \in R\right\}. \tag{2.3}$$

Define

$$d(x, y) = \min_{\sigma\in G} |x - \sigma(y)|, \quad x, y \in \mathbb{R}^N.$$

Recall from [5, Chapter VII, proof of Theorem 2.12] that

$$d(x, y) = |x - y| \text{ if and only if } x, y \in \Gamma \text{ for some chamber } \Gamma. \tag{2.4}$$

Denote by $K_{\text{Dunkl},j}(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [4]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [4, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \cdots, N\}$ and every x, y with $d(x,y) \neq 0$,*

$$|K_{\mathrm{Dunkl},j}(x,y)| \leq \frac{d(x,y)}{|x-y|} \cdot \frac{C}{w(B(x,d(x,y)))}.$$

Proof. The proof is given in [4, Theorem 1.1]. □

Lemma 2.3. *We have*

$$d(x,y)^2 \cdot w(x)w(y) \leq Cw^2(B(x,d(x,y))), \quad x, y \in \mathbb{R}^N.$$

Proof. Define equivalence relation $\sim$ on R by saying $v_1 \sim v_2$ if there exists $g \in G$ with $v_1 = gv_2$. Let us split R into equivalence classes with respect to $\sim$. Since k is G-invariant, it follows that k is constant on each equivalence class. Note that G leaves each equivalence class invariant.

We, therefore, write

$$w(x) = \prod_{A \in R/\sim} \big(\prod_{v \in A} |\langle x, v\rangle| \big)^{k|_A}.$$

On the other hand,

$$w(B(x,r)) \geq c^{(1)}_{R,k} r^N \prod_{A \in R/\sim} \big(\prod_{v \in A} (r + |\langle x, v\rangle|) \big)^{k|_A}.$$

Let $x \in \Gamma$ and $y \in g\Gamma$. Set $z = g^{-1}y \in \Gamma$. Note that $d(x,y) = |x-z|$ and, therefore,

$$|\langle x, v\rangle| + d(x,y) = |\langle x, v\rangle| + |x-z| \geq 2^{-\frac{1}{2}}|\langle z, v\rangle|.$$

Thus,

$$\prod_{v \in A} (|\langle x, v\rangle| + d(x,y)) \geq \prod_{v \in A} 2^{-\frac{1}{2}}|\langle z, v\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle g^{-1}y, v\rangle| =$$

$$= 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, gv\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in gA} |\langle y, v\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v\rangle|.$$

Thus,

$$w(B(x,d(x,y))) \geq c^{(1)}_{R,k} d^N(x,y) \prod_{A \in R/\sim} \big(\prod_{v \in A} (d(x,y) + |\langle x, v\rangle|) \big)^{k|_A} \geq$$

$$\geq c^{(1)}_{R,k} d^N(x,y) \prod_{A \in R/\sim} \big(2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v\rangle| \big)^{k|_A} = c^{(2)}_{R,k} d^N(x,y) w(y).$$

On the other hand,

$$w(B(x,d(x,y))) \geq c^{(1)}_{R,k} d^N(x,y) w(x).$$

Multiplying the last two equations, we complete the proof. □

Lemma 2.4. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\mathrm{Dunkl},j}(x,y) \geq c_{R,k} \frac{x_j - y_j}{|x-y|} \cdot \frac{1}{w(B(x,|x-y|))}.$$

If $x_j \le y_j$, then

$$K_{\mathrm{Dunkl},j}(x,y) \le c_{R,k}\frac{x_j-y_j}{|x-y|}\cdot\frac{1}{w(B(x,|x-y|))}.$$

Proof. The lower bound is identical to that in [4, Theorem 1.2]. □

2.2. **Schatten class.** [Zhijie will add this later]

2.3. **Nearly weakly orthonormal sequences.** Rochberg and Semmes [11] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 7, 10, 1, 2, 9, 11, 13, 12]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [7, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N,\|\cdot\|,w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Definition 2.5. *Let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. A sequence of functions $\{e_Q\}_{Q\in\mathcal{D}}$ is called nearly weakly orthonormal (NWO, in short) sequence if*

(i) $\mathrm{supp}(e_Q)\subset cQ$ *for some positive constant c independent of Q.*
(ii) $\|e_Q\|_{L_\infty(\mathbb{R}^N,w(x)dx)} \le w(Q)^{-\frac{1}{2}}$.

Lemma 2.6. *Let $1<p<\infty$ and let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. If $T\in\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))$, then*

$$\Big\|\Big\{\langle Te_Q,f_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}\Big\}_{Q\in\mathcal{D}}\Big\|_{l_{p,q}(\mathcal{D})} \le c_{N,p,q,w,c_1,c_2}\|T\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N,w(x)dx))}$$

for every NWO systems $\{e_Q\}_{Q\in\mathcal{D}}$ and $\{f_Q\}_{Q\in\mathcal{D}}$ (with respective constants c_1 and c_2).

Proof. reference to [11] is inappropriate. Rochberg and Semmes do not consider weighted setting. we absolutely must provide correct reference. This property can be found in [11, (1.10)] (see also [7, Section 7]). □

3. Comparison of Sobolev norms

Theorem 3.1. *For every $f\in\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p>N$, we have*

$$\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}} \le c_{p,R}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Lemma 3.2. *For every convex polyhedral cone Γ, we have*

$$\int_\Gamma\frac{dx}{|x-y|^{2N}} \ge c_\Gamma \mathrm{dist}(y,\Gamma)^{-N},\quad y\notin\Gamma.$$

Proof. Find a point $z\in\Gamma$ such that

$$|y-z| = \mathrm{dist}(y,\Gamma).$$

We have $z+\Gamma\subset\Gamma$. Thus,

$$\int_\Gamma\frac{dx}{|x-y|^{2N}} \ge \int_{z+\Gamma}\frac{dx}{|x-y|^{2N}} = \int_\Gamma\frac{dx}{|x-(y-z)|^{2N}}.$$

Clearly,

$$|x-(y-z)|\leq |x|+|y-z|.$$

Thus,

$$\int_{\Gamma}\frac{dx}{|x-y|^{2N}}\geq\int_{\Gamma}\frac{dx}{(|x|+|y-z|)^{2N}}=|y-z|^{-N}\int_{\Gamma}\frac{dx}{(|x|+1)^{2N}}.$$

□

Lemma 3.3. *If* $p\in(N,2N)$, *then*

$$\sup_{a>0}\int_{\mathbb{R}^N}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}<\infty.$$

Proof. **Step 1:** we claim that, for every $a>0$,

$$\int_{\mathbb{R}^N}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}\leq$$

$$\leq 4^p\int_{\mathbb{R}^{N-1}\times(-a,a)}\frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}+3^{2N-p}\int_{\mathbb{R}^N}\frac{|x|^{p-2N}dx}{(1+|x|)^p}.$$

We clearly have

$$\int_{\mathbb{R}^N}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}=$$

$$=\int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}+\int_{\mathbb{R}^{N-1}\times\mathbb{R}\backslash(\frac{a}{2},\frac{3a}{2})}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}.$$

If $x_N\notin(\frac{a}{2},\frac{3a}{2})$, then $|x_N-a|\geq\frac{1}{3}|x_N|$. Hence, $|x-ae_N|\geq\frac{1}{3}|x|$. Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}\backslash(\frac{a}{2},\frac{3a}{2})}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}\leq$$

$$\leq\int_{\mathbb{R}^{N-1}\times\mathbb{R}\backslash(\frac{a}{2},\frac{3a}{2})}\frac{3^{2N-p}|x|^{p-2N}dx}{(1+|x|)^p}\leq\int_{\mathbb{R}^N}\frac{3^{2N-p}|x|^{p-2N}dx}{(1+|x|)^p}.$$

If $x_N\in(\frac{a}{2},\frac{3a}{2})$, then (here, we using the notation $x'=(x_1,\cdots,x_{N-1})$)

$$1+|x|\geq 1+\max\{|x'|,\frac{1}{2}a\}\geq\frac{1}{4}(1+|x'|+a).$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})}\frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}\leq 4^p\int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})}\frac{|x-ae_N|^{p-2N}dx}{(1+|x'|+a)^p}=$$

$$=4^p\int_{\mathbb{R}^{N-1}\times(-\frac{a}{2},\frac{a}{2})}\frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}\leq 4^p\int_{\mathbb{R}^{N-1}\times(-a,a)}\frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}.$$

Combining all preceding estimates, we infer the claim in Step 1.

Step 2: we claim that

$$\sup_{a>0}\int_{\mathbb{R}^{N-1}\times(-a,a)}\frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}<\infty.$$

Passing to polar coordinates in $\mathbb{R}^{N-1}$, we equivalently rewrite the claim as

$$\sup_{a>0}\int_{\mathbb{R}_+\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(1+r+a)^p}<\infty.$$

If $0 < a < 1$, then

$$\int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r+a)^p} \le \int_{\mathbb{R}_+\times(0,1)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r)^p}.$$

If $a \ge 1$, then

$$\int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r+a)^p} \le \int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} =$$

$$= \int_{(a,\infty)\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} +$$

$$+ \int_0^a \int_0^r \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dx_N dr}{(r+a)^p} + \int_0^a \int_0^{x_N} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} \le$$

$$\le^{1} \int_{(a,\infty)\times(0,a)} \frac{r^{p-2N} r^{N-2} dr dx_N}{r^p} +$$

$$+ \int_0^a \int_0^r \frac{r^{p-2N} r^{N-2} dx_N dr}{a^p} + \int_0^a \int_0^{x_N} \frac{x_N^{p-2N} r^{N-2} dr dx_N}{a^p} =$$

$$= \int_a^\infty r^{-N-1} dr + a^{-p} \int_0^a r^{p-N-1} dr + \frac{1}{N-1} a^{-p} \int_0^a x_N^{p-N-1} dx_N =$$

$$= \frac{1}{N} a^{-N} + \frac{1}{p-N} a^{-N} + \frac{1}{(N-1)(p-N)} a^{-N} \le \frac{1}{N} + \frac{1}{p-N} + \frac{1}{(N-1)(p-N)}.$$

This yields the claim in Step 2.

Combining Steps 1 and 2, we complete the proof of the lemma. $\square$

Lemma 3.4. *If $p \in (N, 2N)$, then*

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{|x+z|^p} \le c_{p,N} z_N^{-N}, \quad y, z \in \mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Proof. By dilation invariance, we may assume without loss of generality that $z_N = 1$. By translation invariance in $\mathbb{R}^{N-1}$, we may assume without loss of generality that $z_1 = \dots = z_{N-1} = 0$. We, therefore, have

$$|x+z| = |x+e_N| \ge \max\{1, |x|\} \ge \frac{1}{2}(1+|x|), \quad x \in \mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{|x+z|^p} \le 2^p \int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{(1+|x|)^p} \le$$

$$= 2^p \int_{\mathbb{R}^N} \frac{|x-y|^{p-2N} dx}{(1+|x|)^p} = 2^p \int_{\mathbb{R}^N} \frac{\big|x-|y|e_N\big|^{p-2N} dx}{(1+|x|)^p}.$$

The assertion follows from Lemma 3.3. $\square$

We need the following elementary version of the Hahn-Banach theorem.

Lemma 3.5. *Let $K \subset \mathbb{R}^N$ be a convex set and let $y \notin K$. Let $z \in K$ be the point closest to y. Affine hyperplane P passing through z and orthogonal to $y - z$ splits the space into two half-spaces, one of them containing y, the other one containing K.*

[1]It is exactly at this point we are using the assumption $p < 2N$.

Proof. By translation invariance, we may assume without loss of generality that $z=0$. By rotation and dilation invariance, we may assume without loss of generality that $y=e_N$. By assumption, $\mathrm{dist}(e_N,K)=1$. We need to show that K is contained in the (closed) lower half-plane.

Assume the contrary: let $x\in K$ be such that $x_N>0$. Since $0\in K$ and since K is convex, it follows that $\epsilon x\in K$ for every $\epsilon\in(0,1)$. Thus,

$$\mathrm{dist}(e_N,K)\leq|e_N-\epsilon x|,\quad \epsilon\in(0,1).$$

Since $\epsilon\in(0,1)$ is arbitrary, it follows that

$$\mathrm{dist}(e_N,K)\leq\inf_{\epsilon\in(0,1)}|e_N-\epsilon x|=\inf\big(\epsilon^2\sum_{k=1}^{N-1}x_k^2+(1-\epsilon x_N)^2\big)^{\frac12}.$$

Since $x_N>0$, it follows that infimum is strictly less than 1. Thus, $\mathrm{dist}(e_N,K)<1$. However, it is granted in the lemma that $\mathrm{dist}(e_N,K)=1$. This contradiction shows that such an x cannot exist. This completes the proof. $\square$

Lemma 3.6. *If $p\in(N,2N)$, then*

$$\int_\Gamma\frac{|x-y|^{p-2N}dx}{|x-gy|^p}\leq c_p\mathrm{dist}(gy,\Gamma)^{-N},\quad y\in\Gamma.$$

Proof. Find a point $z\in\Gamma$ such that

$$|gy-z|=\mathrm{dist}(gy,\Gamma).$$

By Lemma 3.5, affine hyperplane passing through z and orthogonal to $gy-z$ splits the space into two half-spaces, one of them containing gy, the other one containing Γ (denote this one by H).

We have $y\in H$, $\Gamma\subset H$ and $gy\notin H$. So,

$$\int_\Gamma\frac{|x-y|^{p-2N}dx}{|x-gy|^p}\leq\int_H\frac{|x-y|^{p-2N}dx}{|x-gy|^p}.$$

Using Lemma 3.4, we write

$$\int_H\frac{|x-y|^{p-2N}dx}{|x-gy|^p}\leq c_{p,N}\mathrm{dist}(gy,H)^{-N}.$$

Since $z\in\partial H$ and since $gy-z$ is orthogonal to ∂H, it follows that $\mathrm{dist}(gy,H)=|gy-z|=\mathrm{dist}(gy,\Gamma)$. $\square$

Proof of Theorem 3.1. **Case 1:** Let $p\in(N,2N)$.

We have

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac1p}=$$

$$=\Big(\int_{\Gamma\times\Gamma}\Big(\frac{|x-y|}{|x-gy|}\Big)^p|f(x)-f(gy)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac1p}\leq$$

$$\leq\Big(\int_{\Gamma\times\Gamma}\Big(\frac{|x-y|}{|x-gy|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac1p}+$$

$$+\Big(\int_{\Gamma\times\Gamma}\Big(\frac{|x-y|}{|x-gy|}\Big)^p|f(y)-f(gy)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac1p}\leq$$

$$\leq\Big(\int_{\Gamma\times\Gamma}|f(x)-f(y)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac1p}+$$

$$+\Big(\int_{\Gamma}|f(y)-f(gy)|^p\Big(\int_{\Gamma}\frac{|x-y|^{p-2N}}{|x-gy|^p}dx\Big)dy\Big)^{\frac{1}{p}}.$$

Obviously, the first summand on the right hand side is less than $\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$. By Lemmas 3.6 and 3.2,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+$$

$$+c^{(1)}_{p,R}\Big(\int_{\Gamma}|f(y)-f(gy)|^p\Big(\int_{\Gamma}\frac{dx}{|x-gy|^{2N}}\Big)dy\Big)^{\frac{1}{p}}=$$

$$=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{\frac{1}{p}}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(y)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}+$$

$$+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}.$$

Now,

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}=\Big(\int_{\Gamma\times g\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$$

and

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Thus,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq(1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Summing over Γ and $1\neq g\in G$, we obtain

$$\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq|G|^2(1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Case 2: Let $p\geq 2N$. We have

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^{2N}|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

$\square$

4. Proof of Theorem 1.1 (i)

Let (X,μ) be a measure space and let $L_p(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$ denote the mixed-norm space consisting of all $\mu\times\mu$-measurable functions $f:X\times X\to\mathbb{C}$ such that
$$\int_X \|f(x,\cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x),\ \int_X \|f(\cdot,y)\|_{L_{q,\infty}(X,\mu)}^p d\mu(y)<\infty.$$
We equip $L_p(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$ with its natural norm.

Lemma 4.1. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $K\in L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. The integral operator*
$$T_Kf(x):=\int_X K(x,y)f(y)d\mu(y)$$
belongs to $S_{p,\infty}(L_2(X,d\mu))$. Moreover, we have
$$\|T_K\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|K\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

Proof. The proof is given in [8, Lemma 2] (see also [7, Corollary 11.2]). □

Lemma 4.2. *Let $p>2$ and let $\frac1q=1-\frac2p$. We have*
$$\|F_1F_2\|_{L_p(L_{p',\infty})(X\times X,\mu\times\mu)}\le 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

Proof. We clearly have
$$LHS=\Big(\int_X \|F_1(x,\cdot)F_2(x,\cdot)\|_{L_{p',\infty}(X,\mu)}^p d\mu(x)\Big)^{\frac1p}.$$
Clearly, $\frac{1}{p'}=\frac1p+\frac1q$. Thus,
$$\|F_1(x,\cdot)F_2(x,\cdot)\|_{L_{p',\infty}(X,\mu)}\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_{p,\infty}(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}\le$$
$$\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_p(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}.$$
Hence,
$$LHS\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|_{L_p(X,\mu)}^p\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x)\Big)^{\frac1p}\le$$
$$\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|_{L_p(X,\mu)}^p d\mu(x)\Big)^{\frac1p}\times\sup_{x\in X}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}=$$
$$=2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$
□

Lemma 4.3. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $F\in L_p(X\times X,H^2(\mu\times\mu))$ and $H\in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. We have*
$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|_{L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}^{1-\frac2p}.$$

Proof. By Lemma 4.1,
$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|FH\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$
Let $\frac1q=1-\frac2p$. It follows from Lemma 4.2 that
$$\|F_1F_2\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}\le$$

$$\leq 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

Setting $F_1=FH^{\frac{2}{p}}$ and $F_2=H^{\frac{1}{q}}$, we obtain

$$\|FH\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}\leq$$

$$\leq 2^{\frac{1}{pp'}}\|FH^{\frac{2}{p}}\|_{L_p(X\times X,\mu\times\mu)}\|H^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}=$$

$$=2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{\frac{1}{q}}_{L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}=$$

$$=2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

□

Lemma 4.4. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $F\in L_p(X\times X,GH^2(\mu\times\mu))$ and $H\in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FH}\|_{S_p(L_2(X,d\mu))}\leq c_{p,H}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}.$$

Proof. The assertion follows from Lemma 4.3 by real interpolation. □

Proof of Theorem 1.1 (i). In what follows, let $X=\mathbb{R}^N$ and $d\mu(x)=w(x)dx$. Set

$$F(x,y)=w(B(x,d(x,y)))K_{\mathrm{Dunkl},j}(x,y)(f(x)-f(y)),\quad x\neq y,$$

$$H(x,y)=\frac{1}{w(B(x,d(x,y)))},\quad x\neq y.$$

By [7, Lemma 11.7], $H\in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. By Lemma 2.2,

$$\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\leq$$

$$\leq c^{(1)}_{R,k}\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{w(x)w(y)dxdy}{w(B(x,d(x,y)))^2}\Big)^{\frac{1}{p}}.$$

Using Lemma 2.3, we conclude that

$$\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\leq c^{(2)}_{R,k}\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}.$$

Using Theorem 3.1, we conclude that

$$\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\leq c_{R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator $[R_{\mathrm{Dunkl},j},M_f]$ is FH. The assertion follows now from Lemma 4.4. □

5. Proof of Theorem 1.1 (ii)

Lemma 5.1. *Let $Q\in\mathcal{D}$. We have*

$$K_j(x,y)\geq\frac{c_k}{w(Q)},\quad x\in 3Q,\quad y\in Q+3s(Q)e_j.$$

Proof. TO BE DONE. Follows from Lemma 2.4. □

Lemma 5.2. *Let real-valued $f\in L_1^{\mathrm{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\mathrm{Dunkl},j},M_f]$ is bounded on $L_2(\mathbb{R}^N,w(x)dx)$. For every $Q\in\mathcal{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty,\|\eta_Q\|_\infty\leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)}\int_{3Q}|f(y)-f_{3Q}|w(y)dy\leq c_k|\langle[R_{\mathrm{Dunkl},j},M_f]\xi_Q,\eta_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q+3s(Q)e_j$. Assume for definiteness $\alpha\geq\beta$. Set

$$A_1=\{x\in 3Q:\ f(x)\geq\alpha\},\quad B_1=\{x\in Q+3s(Q)e_j:\ f(x)\leq\beta\},$$
$$A_2=\{x\in 3Q:\ f(x)\leq\beta\},\quad B_2=\{x\in Q+3s(Q)e_j:\ f(x)\geq\beta\},$$
$$\xi_{Q,s}=w(Q)^{-\frac12}\chi_{A_s},\quad \eta_{Q,s}=w(Q)^{-\frac12}\chi_{B_s},\quad s=1,2.$$

We have

$$\langle[R_{\mathrm{Dunkl},j},M_f]\xi_{Q,s},\eta_{Q,s}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}=$$
$$=\frac{1}{w(Q)}\int_{B_s\times A_s}K_j(x,y)(f(y)-f(x))w(x)w(y)dxdy.$$

If $x\in B_1$ and $y\in A_1$, then

$$f(y)-f(x)=(f(y)-\alpha)+(\alpha-\beta)+(\beta-f(x))=$$
$$=|f(y)-\alpha|+|\alpha-\beta|+|f(x)-\beta|\geq|f(y)-\alpha|+|\alpha-\beta|\geq0.$$

If $x\in B_2$ and $y\in A_2$, then

$$f(y)-f(x)=(f(y)-\beta)+(\beta-f(x))=$$
$$=-|f(y)-\beta|-|f(x)-\beta|\leq-|f(y)-\beta|\leq0.$$

Thus,

$$\langle[R_{\mathrm{Dunkl},j},M_f]\xi_{Q,1},\eta_{Q,1}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}\geq$$
$$\geq\frac{1}{w(Q)}\int_{B_1\times A_1}K_j(x,y)\big(|f(y)-\alpha|+|\alpha-\beta|\big)w(x)w(y)dxdy\geq$$
$$\geq\frac{c_k}{w^2(Q)}\int_{B_1\times A_1}\big(|f(y)-\alpha|+|\alpha-\beta|\big)w(x)w(y)dxdy=$$
$$=\frac{c_k}{w^2(Q)}\cdot\Big(w(B_1)\cdot\int_{A_1}|f(y)-\alpha|w(y)dy+|\alpha-\beta|w(B_1)w(A_1)\Big)\geq$$
$$=\frac{c_k}{w^2(Q)}\cdot\Big(\frac12w(Q)\cdot\int_{A_1}|f(y)-\alpha|w(y)dy+|\alpha-\beta|\cdot\frac14w^2(Q)\Big)\geq$$
$$\geq\frac{c_k}{4w(Q)}\cdot\Big(\int_{A_1}|f(y)-\alpha|w(y)dy+|\alpha-\beta|w(Q)\Big)$$

and

$$-\langle[R_{\mathrm{Dunkl},j},M_f]\xi_{Q,2},\eta_{Q,2}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}\geq$$
$$\geq\frac{1}{w(Q)}\int_{B_2\times A_2}K_j(x,y)|f(y)-\beta|w(x)w(y)dxdy\geq$$
$$\geq\frac{c_k}{w^2(Q)}\int_{B_2\times A_2}|f(y)-\beta|w(x)w(y)dxdy=$$
$$=\frac{c_k}{w^2(Q)}\cdot w(B_2)\cdot\int_{A_2}|f(y)-\beta|w(y)dy\geq$$
$$=\frac{c_k}{2w(Q)}\int_{A_2}|f(y)-\beta|w(y)dy.$$

It remains to note that

$$\int_{3Q}|f(y)-\alpha|w(y)dy=\int_{A_1}|f(y)-\alpha|w(y)dy+$$
$$+\int_{A_2}|f(y)-\alpha|w(y)dy+\int_{3Q\setminus(A_1\cup A_2)}|f(y)-\alpha|w(y)dy\leq$$

$$\leq \int_{A_1} |f(y)-\alpha|w(y)dy + \int_{A_2} |f(y)-\beta|w(y)dy$$
$$+\int_{A_2} |\alpha-\beta|w(y)dy + \int_{3Q\setminus(A_1\cup A_2)} |\alpha-\beta|w(y)dy \leq$$
$$\leq \int_{A_1} |f(y)-\alpha|w(y)dy + \int_{A_2} |f(y)-\beta|w(y)dy + 2|\alpha-\beta|w(3Q).$$

Combining the last 3 equations, we conclude

$$\frac{c_k'}{w(Q)}\int_{3Q} |f(y)-\alpha|w(y)dy \leq \max_{s=1,2} |\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_{Q,s}, \eta_{Q,s}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|.$$

If

$$|\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_{Q,1}, \eta_{Q,1}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}| \geq |\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_{Q,2}, \eta_{Q,2}\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c_k'}{w(Q)}\int_{3Q} |f(y)-\alpha|w(y)dy \leq |\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|.$$

Hence,

$$c_k'|f_{3Q}-\alpha| \leq |\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|$$

and, finally,

$$\frac{c_k'}{w(Q)}\int_{3Q} |f(y)-f_{3Q}|w(y)dy \leq 2|\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}|.$$

□

Lemma 5.3. *Suppose that* $1 < p < \infty$ *and* $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$. *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ *for some* $j \in \{1,2,\cdots,N\}$, *then*

$$\Big\|\Big\{\frac{1}{w(3Q)}\int_{3Q} |f(x)-f_{3Q}|w(x)dx\Big\}_{Q\in\mathcal{D}}\Big\|_{l_p(\mathcal{D})} \leq$$
$$\leq c_k\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}.$$

If $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ *for some* $j \in \{1,2,\cdots,N\}$, *then*

$$\Big\|\Big\{\frac{1}{w(3Q)}\int_{3Q} |f(x)-f_{3Q}|w(x)dx\Big\}_{Q\in\mathcal{D}}\Big\|_{l_{N,\infty}(\mathcal{D})} \leq$$
$$\leq c_k\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))}.$$

Proof. We prove only the first assertion. The proof of the second one follows *mutatis mutandi.*

Let $\{\xi_Q\}_{Q\in\mathcal{D}}$ and $\{\eta_Q\}_{Q\in\mathcal{D}}$ be functions constructed in Lemma 5.2. By Lemma 2.6,

$$\Big\{\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N,w(x)dx)}\Big\}_{Q\in\mathcal{D}} \in l_p.$$

The first assertion follows now from Lemma 5.2. □

Lemma 5.4. *Let* Q *be a cube in* $\mathbb{R}^N$ *(with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x\in Q} w(x) \leq \frac{c_k}{m(Q)}\int_Q w(x)dx.$$

Proof. If $x=c_Q+y$, $y\in[-\frac12 s(Q),\frac12 s(Q)]^d$, then

$$|\langle x,v\rangle|\leq|\langle c_Q,v\rangle|+r|\langle y,v\rangle|\leq|\langle c_Q,v\rangle|+\frac{\sqrt{d}}{2}s(Q).$$

Thus,

$$w(x)=\prod_{v\in R}|\langle x,v\rangle|^{k(v)}\leq\prod_{v\in R}(|\langle c_Q,v\rangle|+\frac{\sqrt{d}}{2}s(Q))^{k(v)}.$$

In particular,

$$LHS\leq\prod_{v\in R}(|\langle c_Q,v\rangle|+\frac{\sqrt{d}}{2}s(Q))^{k(v)}.$$

Fix chamber Γ such that $c_Q\in\Gamma$. We have

$$\int_Q w(x)dx\geq\int_{c_Q+(B(0,\frac12 s(Q))\cap\Gamma)}w(x)dx=\int_{B(0,\frac12 s(Q))\cap\Gamma}w(c_Q+y)dy.$$

If $y\in\Gamma$, then

$$|\langle c_Q+y,v\rangle|=|\langle c_Q,v\rangle|+|\langle y,v\rangle|,\quad v\in R.$$

Thus,

$$w(c_Q+y)=\prod_{v\in R}(|\langle c_Q,v\rangle|+|\langle y,v\rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx\geq\int_{B(0,\frac12 s(Q))\cap\Gamma}\prod_{v\in R}(|\langle c_Q,v\rangle|+|\langle y,v\rangle|)^{k(v)}dy.$$

Fix a closed cone $\Gamma'\subset\Gamma$ such that (i) interior of Γ' is non-empty and (ii) $\partial\Gamma\cap\partial\Gamma'=\{0\}$. We have

$$|\langle y,v\rangle|\geq c_{\Gamma'}|y|,\quad y\in\Gamma',\quad v\in R.$$

We, therefore, have

$$\int_Q w(x)dx\geq\int_{B(0,\frac12 s(Q))\cap\Gamma'}\prod_{v\in R}(|\langle c_Q,v\rangle|+c_{\Gamma'}|y|)^{k(v)}dy\geq$$

$$\geq\int_{\substack{y\in\Gamma'\\ \frac14 s(Q)\leq|y|\leq\frac14 s(Q)}}\prod_{v\in R}(|\langle c_Q,v\rangle|+\frac12 c_{\Gamma'}s(Q))^{k(v)}dy=$$

$$=s(Q)^N\prod_{v\in R}(|\langle c_Q,v\rangle|+\frac14 c_{\Gamma'}s(Q))^{k(v)}\times(2^{-N}-4^{-N})\int_{\substack{y\in\Gamma'\\ |y|\leq1}}dy.$$

Combining this with the preceding paragraph, we complete the proof. $\square$

Lemma 5.5. *Suppose $1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}$. Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\Big(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx}\Big)^{\frac1t}\leq c_{t,k}\frac{m(Q)}{\int_Q w(x)dx}.$$

Proof. If $v\in R$ is such that

$$\sqrt{d}s(Q)\leq|\langle c_Q,v\rangle|,$$

then

$$|\langle x,v\rangle|\geq|\langle c_Q,v\rangle|-|\langle x-c_Q,v\rangle|\geq|\langle c_Q,v\rangle|-\frac{\sqrt{d}}{2}s(Q)\geq\frac12|\langle c_Q,v\rangle|,\quad x\in Q.$$

In this case,

$$\int_Q |\langle x,v\rangle|^{(1-t)|R|k(v)}dx \leq 2^{(t-1)|R|k(v)}|\langle c_Q,v\rangle|^{(1-t)|R|k(v)}\cdot\int_Q dx =$$

$$= 2^{(t-1)|R|k(v)}|\langle x_0,v\rangle|^{(1-t)|R|k(v)}\cdot s(Q)^N.$$

If $v\in R$ is such that

$$|\langle c_Q,v\rangle|\leq \sqrt{d}s(Q),$$

then let $y_0=\langle c_Q,v\rangle$ and $z_0=c_Q-y_0v$. Let us introduce new variables $y=\langle x,v\rangle$ and $z=x-yv$ (it varies in the plane orthogonal to v). We write

$$\int_Q |\langle x,v\rangle|^{(1-t)|R|k(v)}dx \leq \int_{|x-c_Q|\leq\frac{\sqrt{d}}{2}s(Q)} |\langle x,v\rangle|^{(1-t)|R|k(v)}dx \leq$$

$$\leq \int_{|y-y_0|,|z-z_0|\leq\frac{\sqrt{d}}{2}r} |y|^{(1-t)|R|k(v)}dydz =$$

$$= (\frac{\sqrt{d}}{2}s(Q))^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y-y_0|\leq\frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)}dy \leq$$

$$= (\frac{\sqrt{d}}{2}s(Q))^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y|\leq\frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)}dy =$$

$$= s(Q)^{N+(1-t)|R|k(v)}\cdot(\frac{\sqrt{d}}{2})^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y|\leq\frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|k(v)}dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x,v\rangle|^{(1-t)|R|k(v)}dx \leq c_1 s(Q)^N\big(\max\{|\langle c_Q,v\rangle|,r\}\big)^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\int_Q w^{1-t}(x)dx \leq \prod_{v\in R}\Big(\int_Q |\langle x,v\rangle|^{(1-t)|R|k(v)}dx\Big)^{\frac{1}{|R|}} \leq$$

$$\leq c_2 s(Q)^N\prod_{v\in R}\big(\max\{|\langle c_Q,v\rangle|,s(Q)\}\big)^{(1-t)k(v)}.$$

The assertion follows easily. □

Lemma 5.6. *Measures dx and $w(x)dx$ (and distance $(x,y)\to|x-y|_\infty$) satisfy the conditions in Proposition 1.2 in* [6].

Proof. That both measures are doubling is obvious.

Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). By Lemmas 5.4 and 5.5,

$$\Big(\frac{1}{m(Q)}\int_Q w^t(x)dx\Big)^{\frac{1}{t}} \leq \frac{c_k}{m(Q)}\int_Q w(x)dx, \quad t>0.$$

$$\Big(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx}\Big)^{\frac{1}{t}} \leq c_G\frac{m(Q)}{\int_Q w(x)dx}, \quad 1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}.$$

The assertion follows now from Proposition 3.6 in [6]. □

Proof of Theorem 1.1 (ii). The left hand side in Lemma 5.3 is the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x,y)\to|x-y|_\infty$. So, the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x,y)\to|x-y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j},M_f]$.

Lemma 5.6 asserts that measures dx and $w(x)dx$ (and distance $(x,y)\to|x-y|_\infty$) satisfy the conditions in Proposition 1.2 in [6]. Appealing to Proposition 1.2 in [6], we conclude that the oscillation semi-norm of the function f with respect to measure dx and distance $(x,y)\to|x-y|_\infty$ is controlled by the one with respect to measure $w(x)dx$ and distance $(x,y)\to|x-y|_\infty$. By the preceding paragraph, the oscillation semi-norm of the function f with respect to measure dx and distance $(x,y)\to|x-y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j},M_f]$.

That oscillation semi-norm of the function f with respect to measure dx and distance $(x,y)\to|x-y|_\infty$ is equivalent to the Sobolev semi-norm of the function f is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [7]. The assertion follows now from the preceding paragraph. □

Proof of Theorem 1.1 (iv). The argument follows the one in the proof of Theorem 1.1 (ii) *mutatis mutandi.* □

6. Intermediate upper-bound results

Theorem 6.1. *Let Γ be a chamber. Let V be an integral operator on $L_2(\Gamma,w(x)dx)$ whose integral kernel K satisfies the condition*

$$|\partial_x^\alpha\partial_y^\beta K(x,y)|\le c_{K,\alpha,\beta}\frac{|x-y|^{-|\alpha|_1-|\beta|_1}}{V(x,|x-y|)},\quad x,y\in\Gamma.$$

We have

$$\|[M_f,V]\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma,w(x)dx))}\le c_K\|f\|_{\dot{W}^{1,N}(\Gamma)}.$$

THE ASSERTION SHOULD FOLLOW FROM HYTONEN.

Theorem 6.2. *Let Γ be a chamber and let $1_G\neq g\in G$. Let V be an integral operator on $L_2(\Gamma,w(x)dx)$ whose integral kernel K satisfies the condition*

$$|\partial_x^\alpha\partial_y^\beta K(x,y)|\le c_{K,\alpha,\beta}\frac{|x-y|^{1-|\alpha|_1-|\beta|_1}}{|x-gy|\cdot V(x,|x-y|)},\quad x,y\in\Gamma.$$

We have

$$\|M_{f-f\circ g}V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma,w(x)dx))}\le c_K(\|f\|_{\dot{W}^{1,N}(\Gamma)}+\|f\circ g\|_{\dot{W}^{1,N}(\Gamma)}).$$

Notation 6.3. *Fix a closed Weyl chamber Γ and let $\Phi:\mathbb{R}^N\to\mathbb{R}^N$ be the linear isomorphism appearing in Corollary* **??**.

Let $\mathbb{D}_\Gamma$ be the dyadic system over Γ constructed in Corollary **??**. *If $I\in\mathbb{D}_\Gamma$ and if $Q=\Phi(I)$, then we denote $d_I=\ell(Q)$.*

Lemma 6.4. *There exists a smooth function ψ supported on $[-1,1]^N\backslash(-\frac14,\frac14)^N$ such that*

$$1=\sum_{I\in\mathbb{D}_\Gamma}\chi_I(x)\chi_{3I}(y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I}).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1,1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\mathrm{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \quad \frac{1}{4} \leq |z|_\infty \leq 1 \right\}. \tag{6.1}$$

For every $z \neq 0$, we have

$$\sum_{j\in\mathbb{Z}} \psi(2^j z) = 1. \tag{6.2}$$

For every $x \in \Gamma$, there exists a unique $I \in \mathbb{D}_{\Gamma,j}$ containing x. For this I, we have $d_I = 2^{-j}$. We have

$$1 = \sum_{j\in\mathbb{Z}} \psi(2^j(\Phi(x) - \Phi(y))) = \sum_{j\in\mathbb{Z}} (\sum_{I\in\mathbb{D}_{\Gamma,j}} \chi_I(x))\psi(2^j(\Phi(x) - \Phi(y))) =$$

$$= \sum_{j\in\mathbb{Z}} \sum_{I\in\mathbb{D}_{\Gamma,j}} \chi_I(x)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}) = \sum_{I\in\mathbb{D}_\Gamma} \chi_I(x)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}).$$

If $x \in I$ and $\psi(\frac{\Phi(x)-\Phi(y)}{d_I}) \neq 0$, then

$$\Phi(x) \in \Phi(I) = Q, \quad |\Phi(x) - \Phi(y)| \leq d_I = \ell(Q).$$

We obviously have $|\Phi(x) - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|\Phi(x) - \Phi(y)|_\infty \leq \ell(Q)$. Thus, $|\Phi(y) - c_Q| \leq \frac{3}{2}\ell(Q)$. In other words, $\Phi(y) \in 3Q$ or, equivalently, $y \in 3I$. Thus,

$$\chi_I(x)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}) = \chi_I(x)\chi_{3I}(y)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}).$$

This completes the proof. □

Lemma 6.5. *Assume that $N > 1$ and $2 < r < \infty$. Let $f \in L_r^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$. There exist functions $\{e_{k,l,I}\}_{k,l\in\mathbb{Z}^N, I\in\mathbb{D}_\Gamma}$, $\{f_{k,l,I}\}_{k,l\in\mathbb{Z}^N, I\in\mathbb{D}_\Gamma}$, such that*

(i) *$e_{k,l,I}$ is supported in I and $f_{k,l,I}$ is supported in $3I$;*

(ii)

$$\|e_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)} = \|f\|_{L_r(\mathbb{R}^N,w(x)dx)}, \quad \|f_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)} \leq c_{K,\kappa} d_I w(I)^{\frac{1}{r}-1};$$

(iii) *For almost every $x, y \in \Gamma$, we have*

$$f(x)K(x,y) = \sum_{k,l\in\mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-2N-1} \sum_{I\in\mathbb{D}_\Gamma} e_{k,l,I}(x) f_{k,l,I}(y). \tag{6.3}$$

Proof. We have

$$K(x,y) = \sum_{I\in\mathbb{D}_\Gamma} \chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}) =$$

$$= \sum_{\substack{I\in\mathbb{D}_\Gamma \\ 3I\subset\Gamma}} \chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}) +$$

$$+ \sum_{\substack{I\in\mathbb{D}_\Gamma \\ 3I\not\subset\Gamma}} \chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x) - \Phi(y)}{d_I}).$$

We will only consider the first sum on the right hand side. The second some can be treated similarly, but the notations become heavier. feel free to write the expression for the second sum if you think it is needed. the only difference is that the domain of the map K_I becomes more complicated.

For every $I\in\mathbb{D}_\Gamma$ with $3I\subset\Gamma$, define the map

$$K_I:[-\frac12,\frac12]^N\times[-\frac32,\frac32]^N\to\mathbb{C}$$

by setting

$$K_I(a,b)=K(c_I+d_I\Phi^{-1}(a),c_I+d_I\Phi^{-1}(b))\psi(a-b),\quad a\in[-\frac12,\frac12]^N,\quad b\in[-\frac32,\frac32]^N.$$

It follows from the assumption on K that feel free to prove this if you think this requires a proof. however, do not prove it here — make a separate lemma instead.

$$\|K_I\|_{C^{4N+2}([-\frac12,\frac12]^N\times[-\frac32,\frac32]^N)}\leq c_K\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)}.$$

Hence, there exists an extension $K_I:\mathbb{T}^N\times\mathbb{T}^N\to\mathbb{C}$ such that

$$\|K_I\|_{C^{4N+2}(\mathbb{T}^N\times\mathbb{T}^N)}\leq c_K'\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)}.$$

In particular,

$$\|K_I\|_{W^{4N+2,2}(\mathbb{T}^N\times\mathbb{T}^N)}\leq c_K''\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)}.$$

We may, therefore, write

$$K_I(a,b)=\sum_{k,l\in\mathbb{Z}^N}\mathrm{coef}_{k,l,I}e^{i\langle k,a\rangle}e^{i\langle l,b\rangle},\quad a,b\in\mathbb{T}^N,$$

where

$$\sum_{k,l\in\mathbb{Z}^N}(1+|k|^2+|l|^2)^{4N+2}|\mathrm{coef}_{k,l,I}|^2\leq(c_K''\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)})^2.$$

In particular,

$$|\mathrm{coef}_{k,l,I}|\leq c_K''(1+|k|^2+|l|^2)^{-2N-1}\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)}.$$

This allows us to write

$$\sum_{\substack{I\in\mathbb{D}_\Gamma\\3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I})=$$

$$=\sum_{\substack{I\in\mathbb{D}_\Gamma\\3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K_I(\frac{\Phi^{-1}(x-c_I)}{d_I},\frac{\Phi^{-1}(x)-c_I}{d_I})=$$

$$=\sum_{\substack{I\in\mathbb{D}_\Gamma\\3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)\sum_{k,l\in\mathbb{Z}^N}\mathrm{coef}_{k,l,I}e^{i\langle k,\frac{\Phi^{-1}(x-c_I)}{d_I}\rangle}e^{i\langle l,\frac{\Phi^{-1}(y-c_I)}{d_I}\rangle}.$$

We now set

$$e_{k,l,I}(x)=f(x)\chi_I(x)e^{i\langle k,\frac{\Phi^{-1}(x-c_I)}{d_I}\rangle},$$

$$f_{k,l,I}(y)=(1+|k|^2+|l|^2)^{2N+1}\mathrm{coef}_{k,l,I}\cdot\chi_{3I}(y)e^{i\langle l,\frac{\Phi^{-1}(y-c_I)}{d_I}\rangle}.$$

In those notations,

$$f(x)\cdot\sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I})=$$

$$=\sum_{k,l\in\mathbb{Z}^N}(1+|k|^2+|l|^2)^{-2N-1}\sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}e_{k,l,I}(x)f_{k,l,I}(y).$$

Obviously,

$$\operatorname{supp}(e_{k,l,I})\subset I,\quad \operatorname{supp}(f_{k,l,I})\subset 3I,$$

$$\|e_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)}=\|f\|_{L_r(I,w(x)dx)},$$

$$\|f_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)}=(1+|k|^2+|l|^2)^{2N+1}|\mathrm{coef}_{I,k,l}|\cdot w(3I)^{\frac1r}\leq$$

$$\leq c_K''\frac{d_I}{(d_I+|c_I-gc_I|)\cdot w(I)}\cdot w(3I)^{\frac1r}\leq c_K'''\frac{d_I}{d_I+|c_I-gc_I|}w(I)^{\frac1r-1}.$$

□

Lemma 6.6. *We have*

$$\|M_fV\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))}\leq c_{K,\kappa}\Big\|\Big\{\frac{d_I}{d_I+|c_I-gc_I|}w(I)^{-\frac1r}\|f\|_{L_r(I,w(x)dx)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}.$$

Lemma 6.7. *For every $I\in\mathbb{D}_\Gamma$, we have*

$$\sup_{x\in I}w(x)\leq c_{\kappa,\Phi}\frac{w(I)}{|I|}.$$

Proof. We have

$$w(I)\approx w(c_I,d_I)\approx d_I^N\prod_{v\in R}(|\langle c_I,v\rangle|+d_I)^{k(v)}.$$

On the other hand, if $x\in I$, then $|x-c_I|\leq c_\Phi d_I$ and, therefore,

$$|\langle x,v\rangle|\leq|\langle x-c_I,v\rangle|+|\langle c_I,v\rangle|\leq|x-c_I|+|\langle c_I,v\rangle|\leq c_\Phi d_I+|\langle c_I,v\rangle|.$$

Thus,

$$w(x)\leq\prod_{v\in R}(c_\Phi d_I+|\langle c_I,v\rangle|)^{k(v)}\leq c_{\Phi,\kappa}\prod_{v\in R}(|\langle c_I,v\rangle|+d_I)^{k(v)}.$$

This completes the proof. □

Lemma 6.8. *Let $1_G\neq g\in G$. We have*

$$\Big\|\Big\{\frac{d_I}{d_I+|c_I-gc_I|}w(I)^{-\frac1r}\|f-f\circ g\|_{L_r(I,w(x)dx)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}\leq$$

$$\leq c_{\kappa,\Phi,r}\Big(\Big\|\Big\{|I|^{-\frac1r}\|f-f_I\|_{L_r(I)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}+$$

$$+\Big\|\Big\{|I|^{-\frac1r}\|f\circ g-(f\circ g)_I\|_{L_r(I)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}+\Big\|\Big\{d_I|I|^{-1}\|H_g\|_{L_1(I)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}\Big).$$

Proof. Denote by $f_{I,w}$ and, respectively, $(f\circ g)_{I,w}$ the averaging (with respect to measure $w(x)dx$) of f (respectively, of $f\circ g$) over I. We, obviously, have

$$\|f-f\circ g\|_{L_r(I,w(x)dx)}\leq\|f-f_{I,w}\|_{L_r(I,w(x)dx)}+$$

$$+\|f\circ g-(f\circ g)_{I,w}\|_{L_r(I,w(x)dx)}+|f_{I,w}-(f\circ g)_{I,w}|\cdot w(I)^{\frac1r}.$$

Obviously,

$$|f_{I,w}-(f\circ g)_{I,w}|\leq\|f-f\circ g\|_{L_1(I,w(x)dx)}\cdot w(I)^{-1}$$

and

$$\|f-f_{I,w}\|_{L_r(I,w(x)dx)}\leq 2\inf_{c\in\mathbb{R}}\|f-c\|_{L_r(I,w(x)dx)}\leq 2\|f-f_I\|_{L_r(I,w(x)dx)},$$

$$\|f\circ g-(f\circ g)_{I,w}\|_{L_r(I,w(x)dx)}\leq 2\inf_{c\in\mathbb{R}}\|f\circ g-c\|_{L_r(I,w(x)dx)}\leq 2\|f\circ g-(f\circ g)_I\|_{L_r(I,w(x)dx)}.$$

Therefore,

$$\frac{d_I}{d_I+|c_I-gc_I|}w(I)^{-\frac1r}\|f-f\circ g\|_{L_r(I,w(x)dx)}\leq$$

$$\leq 2w(I)^{-\frac1r}\|f-f_I\|_{L_r(I,w(x)dx)}+2w(I)^{-\frac1r}\|f\circ g-(f\circ g)_I\|_{L_r(I,w(x)dx)}+$$

$$+\frac{d_I}{d_I+|c_I-gc_I|}w(I)^{-1}\|f-f\circ g\|_{L_1(I,w(x)dx)}.$$

Using Lemma 6.7, we write

$$w(I)^{-\frac1r}\|f-f_I\|_{L_r(I,w(x)dx)}\leq c_{\kappa,\Phi}^{\frac1r}|I|^{-\frac1r}\|f-f_I\|_{L_r(I)},$$

$$w(I)^{-\frac1r}\|f\circ g-(f\circ g)_I\|_{L_r(I,w(x)dx)}\leq c_{\kappa,\Phi}^{\frac1r}|I|^{-\frac1r}\|f\circ g-(f\circ g)_I\|_{L_r(I)}.$$

Obviously,

$$|x-gx|\leq|x-c_I|+|gx-gc_I|+|c_I-gc_I|\leq c'_{\kappa,\Phi}(d_I+|c_I-gc_I|),\quad x\in I.$$

Set

$$H_g(x)=\frac{|f(x)-f(gx)|}{|x-gx|},\quad x\in I.$$

We have

$$\frac{1}{d_I+|c_I-gc_I|}|f-f\circ g|\leq c''_{\kappa,\Phi}H_g.$$

Thus,

$$\frac{1}{d_I+|c_I-gc_I|}\|f-f\circ g\|_{L_1(I,w(x)dx)}\leq c''_{\kappa,\Phi}\|H_g\|_{L_1(I,w(x)dx)}.$$

Using Lemma 6.7, we conclude

$$\frac{1}{d_I+|c_I-gc_I|}w(I)^{-1}\|f-f\circ g\|_{L_1(I,w(x)dx)}\leq c'''_{\kappa,\Phi}\frac{1}{|I|}\|H_g\|_{L_1(I)}.$$

Combining these estimates, we complete the proof. □

7. Constant characterization

Theorem 7.1. *If $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$ is such that*

$$[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$$

for some $p \le d$, then f is constant.

Proof. By Lemma 5.6, measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_\Gamma}$ satisfy on Γ the assumptions in Theorem 1.4 in [6]. By assumption, $[T_{\Gamma,j}, M_f] \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$. By Theorem 1.4 in [6], f is constant on Γ.

Hence, f is constant on every chamber. Let $f = c_1$ on Γ_1 and $f = c_2$ on Γ_2. On the other hand, $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{2N}(L_2(\mathbb{R}^N, w(x)dx))$. By Theorem 1.1 (ii), $f \in \dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)$. We have

$$\int_{\Gamma_1 \times \Gamma_2} \frac{|c_1 - c_2|^{2N} dxdy}{|x-y|^{2N}} = \int_{\Gamma_1 \times \Gamma_2} \frac{|f(x) - f(y)|^{2N} dxdy}{|x-y|^{2N}} \le$$
$$\le \int_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|f(x) - f(y)|^{2N} dxdy}{|x-y|^{2N}} = \|f\|_{\dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)} < \infty.$$

It follows that $c_1 = c_2$. Since chambers Γ_1 and Γ_2 are arbitrary, the assertion follows. □

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References

[1] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. 5

[2] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. 5

[3] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at https://arxiv.org/abs/2403.08249, 2024. 5

[4] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. 3, 4, 5

[5] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. 3

[6] T. Hytonen. A_∞-invariance of oscillatory norms and schatten characterisations of commutators. 2, 15, 16, 21

[7] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at https://arxiv.org/abs/2411.02613, 2024. 5, 10, 11, 16

[8] S. Janson and T.H. Wolff. Schatten classes and commutators of singular integral operators. *Ark. Mat.*, 20(2):301–310, 1982. 10

[9] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. 5

[10] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. 5

[11] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. 5

[12] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at https://arxiv.org/abs/2411.05810, 2024. 5

[13] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at https://arxiv.org/abs/2407.14988, 2024. 5

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